Introduction to Probability and Inference

September 28, 2022

As you come in...

1. Open an RScript, and save it as "ProbabilityPractice.R" in your PLSC 309 folder.
2. Take a die (a few will be passed around). Roll the die on your table 15 times, and record the result in a vector object in R:
   - For instance: rolls$<-$ c(3,4,1,2....)
3. Calculate the mean of your rolls and save that number as an object called "mean_rolls"
4. Calculate the median as well...
5. Calculate the standard deviation as well...
6. Create a histogram of your "rolls" data. Add a proper title and x-axis label. Make the color "chocolate3" (yum)
7. Create a boxplot of the same thing, add a proper title, and make the color "cyan3"

Terminology

Population- the data for every possible relevant case

Sample- a subset of cases from an underlying population (ideally randomly drawn)

Statistical Inference- using what we know from our sample to make a claim about the population, but *with some reasonable degree of uncertainty*.

- Sample size and certainty level?

Probability I

Outcome- an observation of some event

Independent Outcomes- when the realization of one outcome does not affect the realization of the other outcome(s)

- dice, flipping coins, etc.

Basic Law of Probability

Given that all possible outcomes of an event are equally likely, the probability of any specified outcome is equal to the number of ways that the outcome could be achieved divided by the total number of ways that all possible outcomes could be achieved.

Some Features of Probability

1. Always range between 0 and 1
   - Probability of Heads ....
2. Sum of probabilities for all possible outcomes is 1
3. Probability of Outcome A <u>NOT</u> happening is 1 - P(A)
4. Probability of Outcome A <u>OR</u> Outcome B is P(A) + P(B)
5. Probability of Independent Outcome A <u>AND</u> Independent Outcome B is P(A) x P(B)

Probability Distribution of Six-sided Dice Rolls

What is the probability of rolling a 1?

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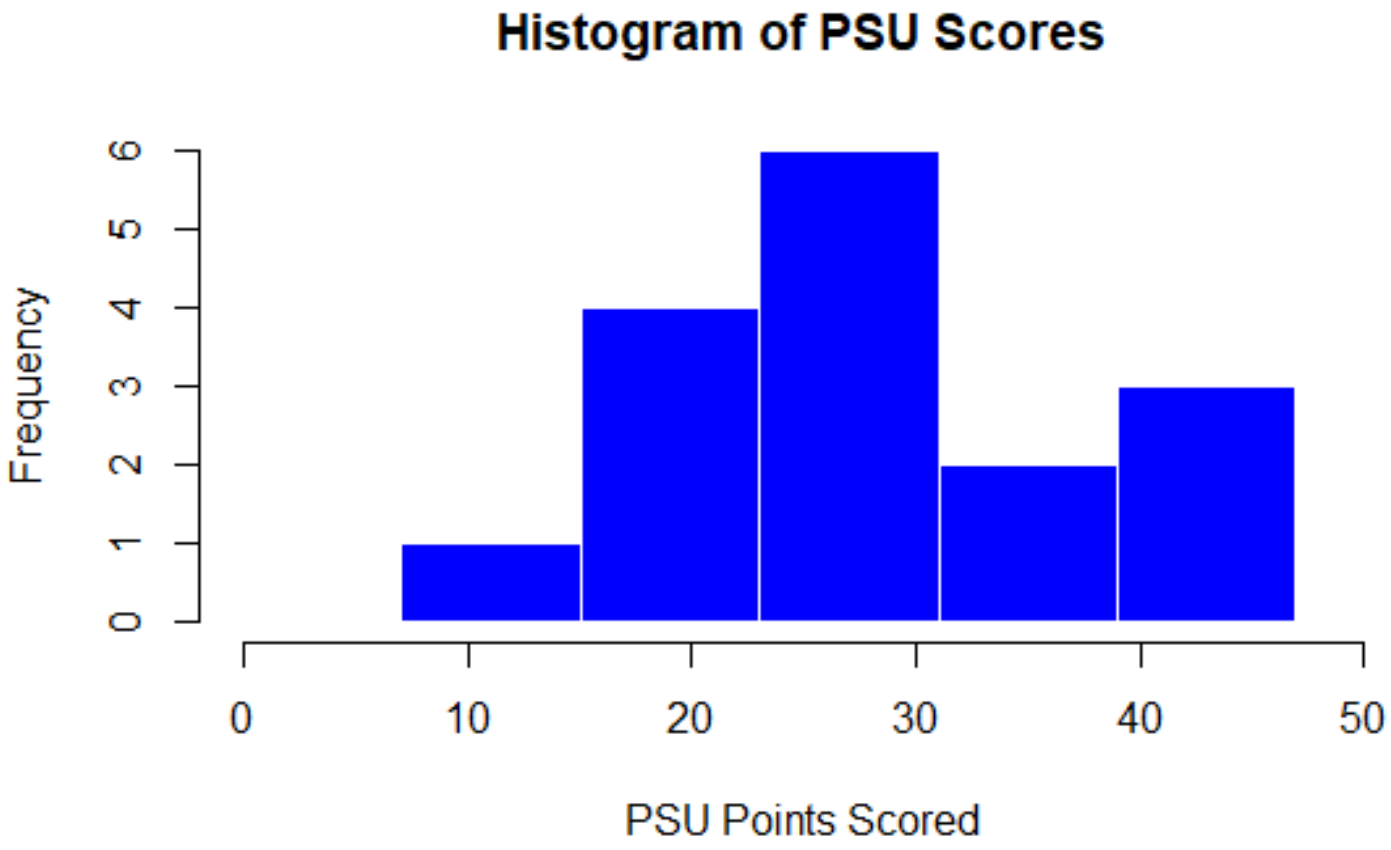
What is the probability of rolling a 1 and a 2 if you roll two dice at the same time?

Vegas! (And Why I Lost So Much Money)

Frequency Distribution (Binned)

Range of Points Scored	**Frequency of Games**
0-7	0
8-15	1
16-23	4
24-31	6
32-39	3
40-47	2
Total Games=	16

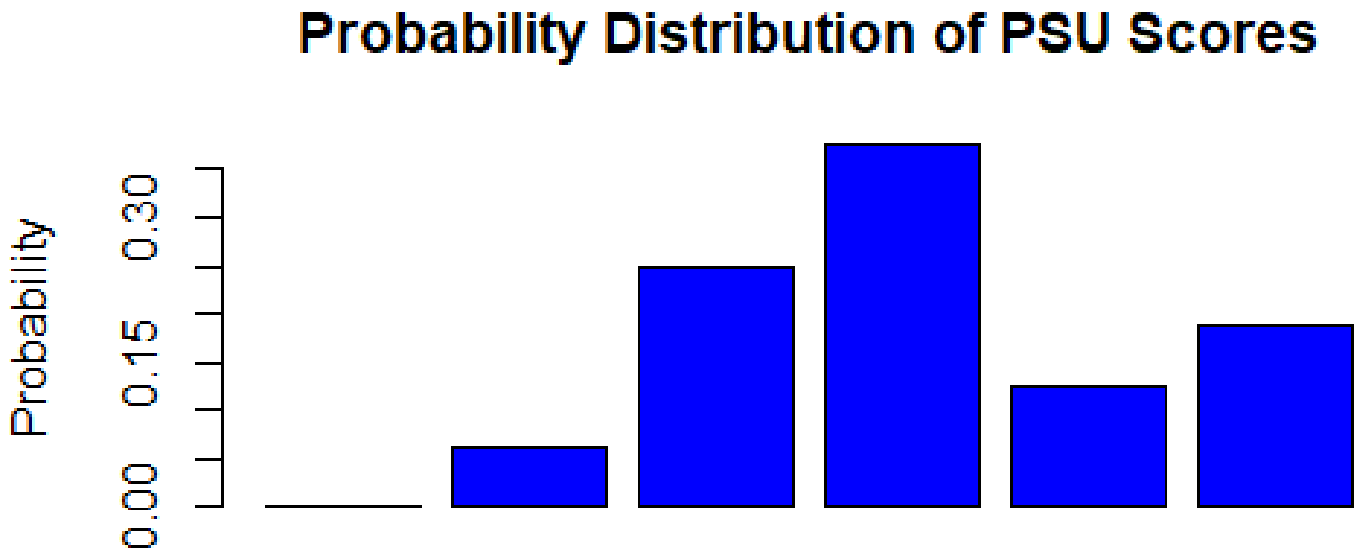
Histograms


Histogram of PSU Scores
Frequency
0
1
2
3
4
5
6
0
10
20
30
40
50
PSU Points Scored

Probability Distribution

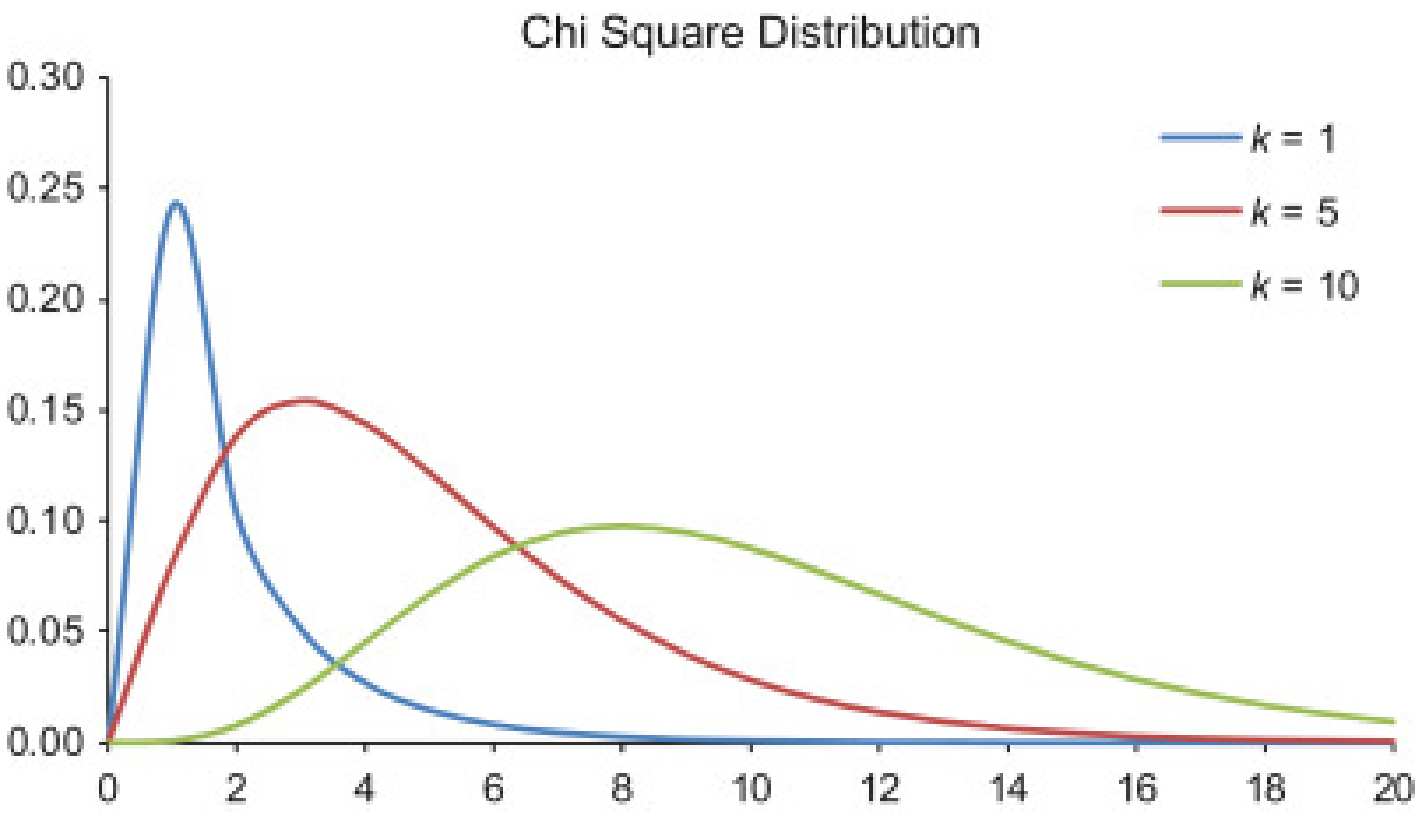
Range of Points Scored	Frequency of Games	Probability
0-7	0	0
8-15	1	0.0625
16-23	4	0.25
24-31	6	0.375
32-39	3	0.1875
40-47	2	0.125
Total Games=	16	100%

Probability Distribution (Visualized)

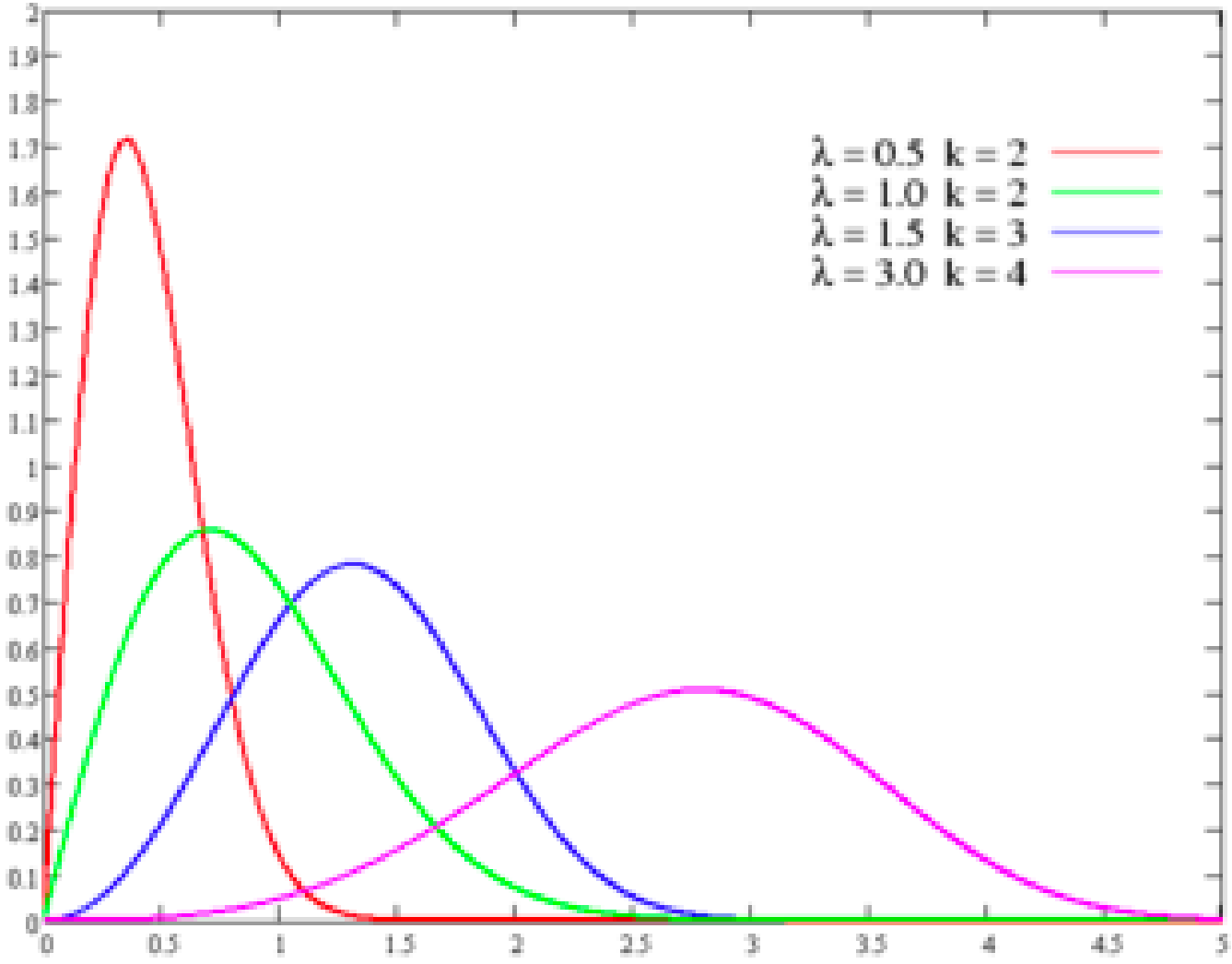


Shows the distributions of (observed) probabilities for each event

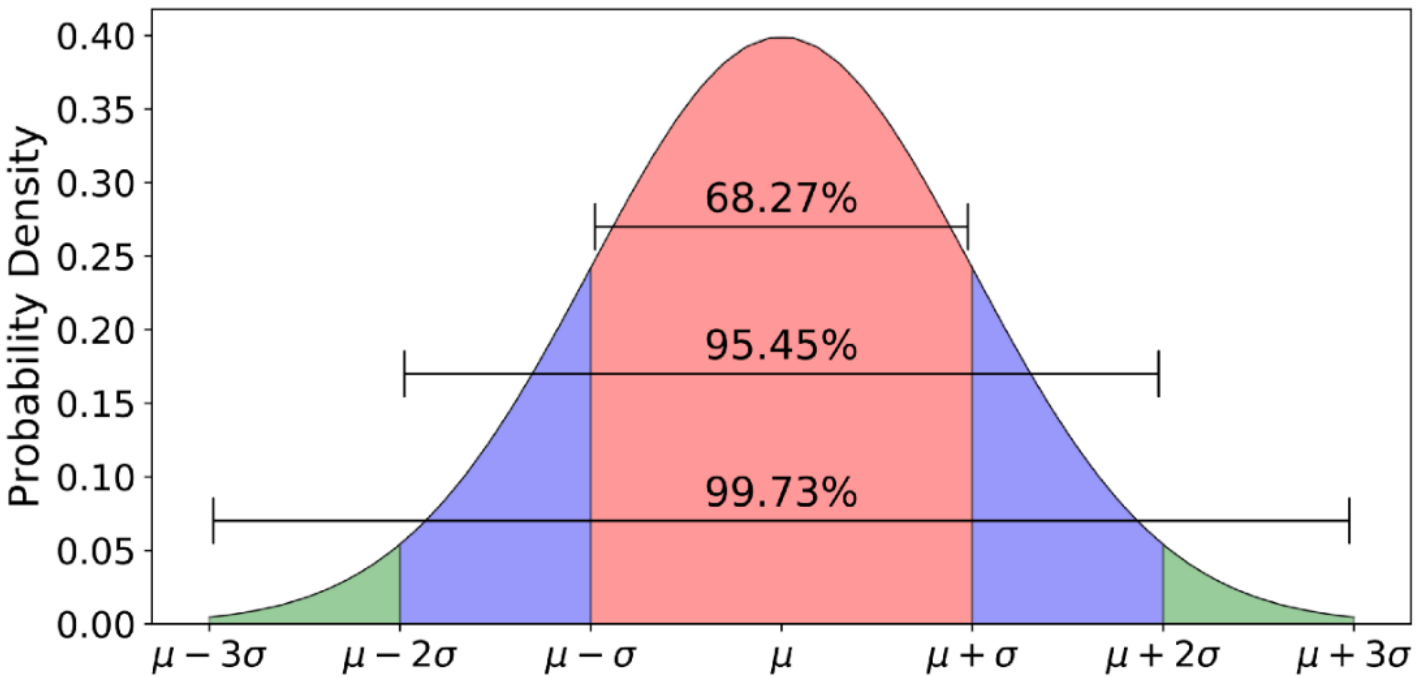
Theoretical Distribution: Chi Square



Theoretical Distribution: Weibull



The Normal Distribution



$\mu =$ Mean

$\sigma =$ standard deviation

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For example:

- You each made a distribution of dice rolls.
- A sampling distribution would be the distribution of the means of all of your individual distributions.

Sampling Distributions

Let's do it...

Central Limit Theorem

The mind-blowing math thing...

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2. The sampling distribution would be normally distributed around the mean
3. The standard deviation of the sampling distribution is $\sigma_{\bar{Y}} = \frac{s_Y}{\sqrt{n}}$
4. And this would be true *even if the distributions of each of the samples is not normal*

Why Standard Deviation is Important (Intro)

Recall: the standard deviation of a distribution is basically the average deviation of all values of x from the mean.

One of two parameters that defines the shape of the normal (and many other) distributions

The <u>standard</u> deviation is a <u>standardized</u> measure is the foundation for calculating *z-scores*

Next Class

Confidence Intervals and One-sample Tests

- Due: Data Viz Lab Assignment
- Tutorials from "Seeing Theory"
- Don't worry about the "submit confidence interval calculations" point